

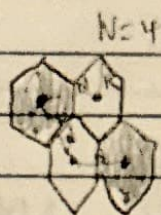
HW 7

1. Consider a cellular system that employs the N -cell cluster size w/ hex geo.

D = min dist between centers of cells that use same freq band

R = radius of cell

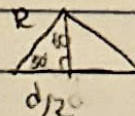
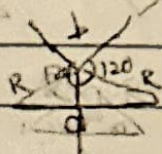
d = dist between adj cell centers



$N=4$

$$D = 2d, N=4, D = \sqrt{4}d, \boxed{D = \sqrt{N}d}$$

Hexagon angle is 120°



$$\begin{aligned} d/2 &= R \sin 60^\circ \\ R &= 2d \sin 60^\circ \\ d &= R \sin 60^\circ \end{aligned}$$

Then b/c $D = \sqrt{N} \cdot d$ and $d = R \sin 60^\circ$, $\boxed{D = \sqrt{3N} \cdot R}$

2. Just refer to the HW doc, way too long to write.

$$P_r = P_t (\lambda / 4\pi d)^2 \quad P = \sqrt{2} R$$

Find signal to interference ratio (SIR) at the MS in dB for the worst case scenario

$$BS \quad P_r = P_T (\lambda / 4\pi R)^2$$

$$\text{otherwise: } P_{r1} = P_T (\lambda / 4\pi (D+R))^2$$

$$P_{r2} = P_T (\lambda / 4\pi (D-R))^2$$

$$P_{r3} = P_T (\lambda / 4\pi (D))^2$$

$$P_{r4} = P_T (\lambda / 4\pi (D+R))^2$$

$$P_{r5} = P_T (\lambda / 4\pi (D+R))^2$$

$$P_{r6} = P_T (\lambda / 4\pi (D))^2$$

$$SIR = \frac{P_r}{P_{r1} + P_{r2} + P_{r3} + P_{r4} + P_{r5} + P_{r6}}$$

$$= \frac{\frac{1}{(R^2)}}{\frac{2}{(D+R)^2} + \frac{2}{D^2} + \frac{2}{(D+R)^2}} \quad D = \sqrt{21} R$$

$$= \frac{\frac{1}{R^2}}{\frac{2}{(R\sqrt{21}+R)^2} + \frac{2}{(R\sqrt{21})^2} + \frac{2}{(R\sqrt{21}+R)^2}}$$

$$= \frac{\frac{1}{R^2}}{\frac{2}{R^2(\sqrt{21}-1)^2} + \frac{2}{R^2(21)} + \frac{2}{R^2(\sqrt{21}+1)^2}}$$

$$= \frac{1}{\frac{2}{(\sqrt{21}-1)^2} + \frac{2}{21} + \frac{2}{(\sqrt{21}+1)^2}} = \frac{1}{3.15} = 3.172$$

$$10 \log_{10}(3.172) = \boxed{5.014 \text{ dB}}$$

3. 4 cellular systems. Freq bands: 825-845 MHz for mobile
870-890 MHz for base station

one 30 kHz channel in each direction,

Systems are distinguished w/ reuse factors: 4, 7, 12, 19

- a. In each system, the cluster of cells (4, 7, 12, 19) is duplicated 16 times. # of simultaneous calls supported by each system?

$$845 - 825 = 20 \text{ MHz} \quad 890 - 870 = 20 \text{ MHz}$$

$$20 + 20 = 40 \text{ MHz}$$

duplex channel =

simultaneous

calls

$$30 \times 2 = 60 \text{ kHz for two way}$$

$$\frac{40 \times 10^6 (16)}{60 \times 10^3} = 10666.67 \text{ (must round down)}$$

10666 channels

- b. Find # of simultaneous calls supported by a single cell in each system.

$$\frac{40 \times 10^6}{60 \times 10^3} = 666.67 \text{ calls total} \rightarrow \text{round to } 666$$

$$N=4: 666/4 = 166.5 \rightarrow 166 \text{ channels}$$

$$N=7: 666/7 = 95.143 \rightarrow 95 \text{ channels}$$

$$N=12: 666/12 = 55.5 \rightarrow 55 \text{ channels}$$

$$N=19: 666/19 = 35.053 \rightarrow 35 \text{ channels}$$

c. area covered in cells in each system?

$$N=4: 4 \times 16 = 64 \text{ cells}$$

$$N=7: 7 \times 16 = 112 \text{ cells}$$

$$N=12: 12 \times 16 = 192 \text{ cells}$$

$$N=19: 19 \times 16 = 304 \text{ cells}$$

d. Suppose cell size is same in all 4 systems w/ fixed area 100 cells, # of simultaneous comms in each system?

$$N=4, 166 \times 100 = 16,600 \text{ channels}$$

$$N=7, 95 \times 100 = 9,500 \text{ channels}$$

$$N=12, 55 \times 100 = 5,500 \text{ channels}$$

$$N=19, 35 \times 100 = 3,500 \text{ channels}$$

4. For IEEE 802.11n, determine max data rate for 16 QAM using a 2/3 coding rate w/ 2 parallel data streams in these cases

a. 20 MHz channel (48 subcarriers carry data)

$$\text{Data rate} = \frac{(\# \text{ of data subcarriers}) (\text{bits per subcarrier}) (\text{coding rate}) \times (\# \text{ of spatial streams})}{\text{OFDM symbol time}}$$

$3.6 \times 10^6 \approx$ start spread intensity
for 802.11n.

$$\frac{(48) * (\log_2(16)) (2/3) (2)}{3.6 \times 10^6} = \boxed{71.7 \text{ Mb/s}}$$

4. 40 MHz (108 subcarriers)

$$\frac{(108) * \log_2(16) (2/3) (2)}{3.6 \times 10^6} = \boxed{160 \text{ Mb/s}}$$

5. Consider OFDM parameters

$$f_c \approx 1 \text{ GHz}$$

$$v_{\text{max}} \approx 51.94 \text{ m/s}$$

$$t_{\text{max delay}} = 4 \mu\text{s}$$

16QAM mod

$$\text{coding rate} = 5/6$$

$$\text{Net data rate } R \approx 50 \text{ Mb/s}$$

a. min CP to avoid ISI?

$$\boxed{T_c = 4 \mu\text{s}}$$

b. Find doppler spread f_D

$$f_D \approx \Delta f \frac{v}{c} \approx 1 \times 10^9 \left(\frac{51.94}{3 \times 10^8} \right) \approx \boxed{173.148 \text{ Hz}}$$

c. SIR can be approximated (linear units) as follows

$$SIR \approx \left\{ \frac{12}{(2\pi f_D T_s)^2} \right\} \quad \begin{array}{l} f_D = \text{doppler spread} \\ T_s = \text{symbol duration} \end{array}$$

Find max T_s (round down to nearest ms) so $SIR \geq 30\text{dB}$

$$SIR \geq 30\text{dB} \rightarrow 10 \log_{10} \left(\frac{12}{(2\pi f_D T_s)^2} \right) \geq 30\text{dB}$$

$$\frac{12}{(\pi(123.148)T_s)^2} \geq 10^{\frac{30}{10}}$$

$$\frac{1}{T_s^2} \geq 99631171.73$$

$$1.0069 \times 10^4 \geq T_s$$

$$T_s \leq 100\text{ms}$$

$$\text{max } T_s = 100\text{ms}$$

d. Determine the subcarrier spacing Δf , # of data subcarriers N , & BW to support data Rate $R = 50\text{Mb/s}$

$$T_s = 100\text{ms}$$

$$\log_2(16) = 4$$

coding rate

$$50 \times 10^6 = N \left(\frac{4(1/6)}{100 \times 10^6 + 4 \times 10^6} \right)$$

$$N \approx 1560 \text{ subcarriers}$$

$$\Delta f = \frac{1}{T_s} = \frac{1}{100 \times 10^{-6}} = 10\text{kHz}$$

$$BW \geq N \cdot \Delta f = 1560 \cdot 10 \times 10^3 = 15.6\text{MHz}$$